DLOPs Assignment-1

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Q1(a). Train a deep learning model on the MNIST and FashionMNIST datasets with 70%-10%-20% train-val-test split. Use the Following Models: ResNet-18 and ResNet-50. Fill in the classification accuracy in the table below and provide a detailed analysis based on the results.

Note: Also perform experiments by varying hyperparameters.
pin_memory=False,True
Epochs = vary the number of epochs by yourself (perform an experiment with any two total numbers of epochs).
Keep these factors Constant:- USE_AMP=True
The valid experiments include achieving a Classification accuracy of more than 80+%

LINK: https://colab.research.google.com/drive/1wcajqES_-zzqdo2eFKQkOE2NJfJflfuI

For MNIST :-

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**	96.97	95.01
16	**SGD**	**0.0001**	63.48	40.31
16	**Adam**	**0.001**	98.94	98.87
16	**Adam**	**0.0001**	98.95	99.21
32(Optional)	**SGD**	**0.001**	98.93	98.36
32(Optional)	**SGD**	**0.0001**	97.59	95.02
32(Optional)	**Adam**	**0.001**	98.72	98.15
32(Optional)	**Adam**	**0.0001**	98.56	96.71

For Fashion MNIST :-

			Test Classification Accuracy (in %)	
Batch Size	**Optimizers**	**Learning Rate**	**ResNet-18**	**ResNet-50**
16	**SGD**	**0.001**	81.59	75.48
16	**SGD**	**0.0001**	65.08	54.05
16	**Adam**	**0.001**	89.75	88.99
16	**Adam**	**0.0001**	91.97	90.64
32	**SGD**	**0.001**	88.38	86.82
32	**SGD**	**0.0001**	87.74	80.21
32	**Adam**	**0.001**	89.39	87.04
32	**Adam**	**0.0001**	89.16	85.12

ResNet-18 and ResNet-50 achieved very high accuracy on MNIST and good performance on FashionMNIST. The Adam optimizer gave the best results, while SGD with low learning rate performed poorly. ResNet-18 slightly outperformed ResNet-50, showing that a simpler model is sufficient for these datasets. Batch size 32 gave slightly better results than 16.

Q1(b). Also train a SVM classifier on MNIST and FashionMNIST datasets with varying SVM hyperparameters with kernels such as: ‘poly’, ‘rbf’. Report Testing Classification Accuracy with training time (in ms).

LINK: https://colab.research.google.com/drive/1inGC5FWQ-8Ni7MOfbd3l-a6MY0uJD0bO

Dataset Name	Kernel	Accuracy	Time_ms
MNIST	**Poly**	**0.93**	**8223**
MNIST	**rbf**	**0.94**	**8356**
Fashion MNIST	**Poly**	**0.82**	**8725**
Fashion MNIST	**rbf**	**0.86**	**8121**

SVM classifiers with RBF kernel performed better than Polynomial kernel on both MNIST and FashionMNIST. MNIST achieved higher accuracy than FashionMNIST, showing that FashionMNIST is more challenging. RBF also achieved better accuracy with comparable training time, making it the better choice for this task.

Q2. For the **FashionMNIST** dataset, report performance and train-time based on running the experiments on CPU and GPU both. Write a detailed report analyzing the training time, number of FLOPs used and the classification accuracy of the models.

LINK:
https://colab.research.google.com/drive/1NYub6BculXzy1_In5vJ5aDkCqnSMslsj#scrollTo=sLqDM6P8B6FP

				Test Classification Accuracy (in %)			**Train Time (in ms)**			**FLOPs**		
Compute	**Batch Size**	**Optimizer**	**Learning Rate**	**ResNet-18**	**ResNet-32 (Optional)**	**ResNet-50**	**ResNet-18**	**ResNet-32 (Optional)**	**ResNet-50**	**ResNet-18**	**ResNet-32 (Optional)**	**ResNet-50**
CPU	**16**	**SGD**	**0.001**	46.25		23.05	9.511200e+05		2.424643e+06	1824805898		4130408458
CPU	**16**	**Adam**	**0.001**	75.60		70.35	9.891068e+05		2.355874e+06	1824805859		4130408458
GPU	**16**	**SGD**	**0.001**	35.55		17.80	1.711225e+04		3.268896e+04	1824805898		4130408458
GPU	**16**	**Adam**	**0.001**	72.45		65.70	1.578898e+04		3.300805e+04	1824805898		4130408458

ResNet-18 and ResNet-50 were trained on FashionMNIST using CPU and GPU. Adam achieved higher accuracy than SGD, and ResNet-18 performed better and was more efficient than ResNet-50. GPU was much faster than CPU while giving similar accuracy. A subset of the dataset was used to reduce training time while preserving performance trends.